

IT CREATIVE ACADEMY

07 - DARS

Loyiha modelini tuzish

Django ORM · Model · ForeignKey · Migration · Admin

ITC-Blog loyihasi uchun amaliy qo'llanma
Django 4.2 · Python 3.10

Dars haqida

Ushbu darsda biz **06-darsda** chizgan ma'lumotlar bazasi (database) chizmasini amaliyotga aylantiramiz. Ya'ni ITC-Blog loyihasi uchun to'liq, professional **model** tuzamiz. Bitta katta jadval o'rniga bir nechta o'zaro bog'langan modellar yaratib, ma'lumotlarni to'g'ri tashkil qilishni o'rganamiz.

Dars yakunida siz quyidagilarni bilasiz:

- Model nima va u qanday qilib jadvalga aylanadi;
- Bir nechta modelni **ForeignKey** orqali bog'lash;
- To'g'ri field (maydon) turlarini tanlash;
- **makemigrations** va **migrate** buyruqlari farqi;
- Modellar admin panelga chiroyli qilib ulash.

1. Takrorlash: 06-darsda nima qilgandik?

06-darsda barcha ma'lumotni bitta News modeliga joylagandik. Bu "yassi" (flat) model deyiladi:

```
class News(models.Model):
    title = models.CharField(max_length=200)
    content = models.TextField()
    file = models.FileField(upload_to='news_file/')
    likes = models.IntegerField(default=0)
    comments = models.IntegerField(default=0) # <-- faqat son!
    views = models.IntegerField(default=0)
    tags = models.CharField(max_length=100) # <-- matn ko'rinishida
```

Muammo nimada?

- **comments** faqat son — kim, qachon, nima yozganini bilmaymiz;
- **tags** oddiy matn — bo'limlar bo'yicha filtr qilib bo'lmaydi;
- Maqola muallifi (author) yo'q;
- Ma'lumotlar takrorlanadi va boshqarish qiyin.

2. Yechim: bog'lanishli (relational) model

Yechim — ma'lumotni mantiqiy bo'laklarga ajratish va ularni **kalit (key)** orqali bog'lash. Biz uchta model tuzamiz: **Category** (bo'lim), **News** (yangilik) va **Comment** (izoh).

ITC-Blog ma'lumotlar bazasi chizmasi (ER-diagramma):



Bog'lanish (1 - N) nimani anglatadi?

- Bitta **Category** ko'plab **News**ga ega bo'lishi mumkin;
- Bitta **News** ko'plab **Comment**ga ega bo'lishi mumkin;
- Bunday bog'lanish Django'da **ForeignKey** bilan yoziladi.

3. Field (maydon) turlari

Modelda ko'p ishlatiladigan asosiy field turlari:

Field turi	Vazifasi
CharField	Qisqa matn (sarlavha, nom). max_length majburiy.
TextField	Uzun matn (maqola tanasi, izoh).
SlugField	URL uchun mos matn: 'yangi-maqola'.
ImageField	Rasm yuklash (Pillow kutubxonasi kerak).
IntegerField	Butun son. PositiveIntegerField — manfiy emas.
DateTimeField	Sana va vaqt (auto_now_add, auto_now).
ForeignKey	Boshqa modelga bog'lash (1 - N aloqa).

4. models.py — modellarni yozamiz

Endi news/models.py faylini to'ldiramiz:

```
from django.db import models
from django.contrib.auth.models import User

class Category(models.Model):
    name = models.CharField(max_length=100)
    slug = models.SlugField(max_length=120, unique=True)
    created_at = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return self.name

class News(models.Model):
    class Status(models.TextChoices):
        DRAFT = 'draft', 'Qoralama'
        PUBLISHED = 'published', 'Chop etilgan'

    title = models.CharField(max_length=200)
    slug = models.SlugField(max_length=220, unique=True)
    category = models.ForeignKey(Category,
        on_delete=models.CASCADE, related_name='news')
    author = models.ForeignKey(User,
        on_delete=models.CASCADE, related_name='news')
    image = models.ImageField(upload_to='news/images/',
        blank=True, null=True)
    content = models.TextField()
    status = models.CharField(max_length=10,
        choices=Status.choices, default=Status.DRAFT)
    views = models.PositiveIntegerField(default=0)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    def __str__(self):
        return self.title

class Comment(models.Model):
    news = models.ForeignKey(News,
        on_delete=models.CASCADE, related_name='comments')
    user = models.ForeignKey(User,
        on_delete=models.CASCADE, related_name='comments')
    body = models.TextField()
    created_at = models.DateTimeField(auto_now_add=True)
```

5. Muhim tushunchalarni eslab qoling

Tushuncha	Ma'nosi
ForeignKey	Bitta modelni boshqasiga bog'laydi. Masalan har bir News qaysidir Category'ga tegishli.
on_delete=CASCADE	Bog'langan obyekt o'chsa, unga tegishlilari ham o'chadi. Masalan Category o'chsa, undagi News ham o'chadi.
related_name	Teskari bog'lanish nomi: category.news.all() — shu bo'limdagi barcha yangiliklar.
__str__	Obyekt admin panelda va konsolda qanday ko'rinishini belgilaydi.
auto_now_add / auto_now	Birinchisi — yaratilgan vaqtni bir marta yozadi; ikkinchisi — har saqlashda yangilanadi.
TextChoices	Tayyor variantlardan tanlash (Qoralama / Chop etilgan).

6. Migration: modelni bazaga o'tkazamiz

Model — bu Python kodi. Uni real jadvalga aylantirish uchun ikki buyruq ishlatamiz:

```
# 1) O'zgarishlardan migration fayli tayyorlaydi
python manage.py makemigrations

# 2) Migration'larni bazaga (db.sqlite3) qo'llaydi
python manage.py migrate
```

Ikkisining farqi:

- **makemigrations** — "reja" tuzadi (migrations/ papkasiga fayl yozadi);
- **migrate** — rejani bajaradi (bazada jadval yaratadi).

7. Admin panelga ulash

news/admin.py faylida modellarni ro'yxatdan o'tkazamiz:

```
from django.contrib import admin
from .models import Category, News, Comment

@admin.register(News)
class NewsAdmin(admin.ModelAdmin):
    list_display = ('title', 'category', 'author',
                   'status', 'views', 'created_at')
    list_filter = ('status', 'category')
    search_fields = ('title', 'content')
    prepopulated_fields = {'slug': ('title',)}

admin.site.register(Category)
admin.site.register(Comment)
```

Admin foydalanuvchisini yaratib, panelni ochib ko'ring:

```
python manage.py createsuperuser  
python manage.py runserver  
# Brauzer: http://127.0.0.1:8000/admin/
```

8. Amaliy topshiriq (uy vazifasi)

Mustaqil bajaring:

1. Yuqoridagi uchta modelni o'z loyihangizda yozing;
2. **makemigrations** va **migrate** buyruqlarini ishga tushiring;
3. **createsuperuser** orqali admin yarating;
4. Admin panel orqali 2 ta Category va 3 ta News qo'shing;
5. Qo'shimcha: **Comment** modeliga 'likes' (PositiveIntegerField) qo'shib ko'ring.

Xulosa

Bu darsda biz yassi modelni professional, bog'lanishli modelga aylantirdik. Endi loyihamizda **Category**, **News** va **Comment** jadvallari o'zaro **ForeignKey** orqali bog'langan. Keyingi darsda ushbu modellardan foydalanib, ma'lumotlarni saytda ko'rsatishni o'rganamiz.

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